UNIVERSAL ROBOTS

Company	Universal Robots
Drawing number	3.0.0.2
Project name	Universal Robots UR10
Created on	25-09-2014
Edit date	24-11-2015
Number of pages	19

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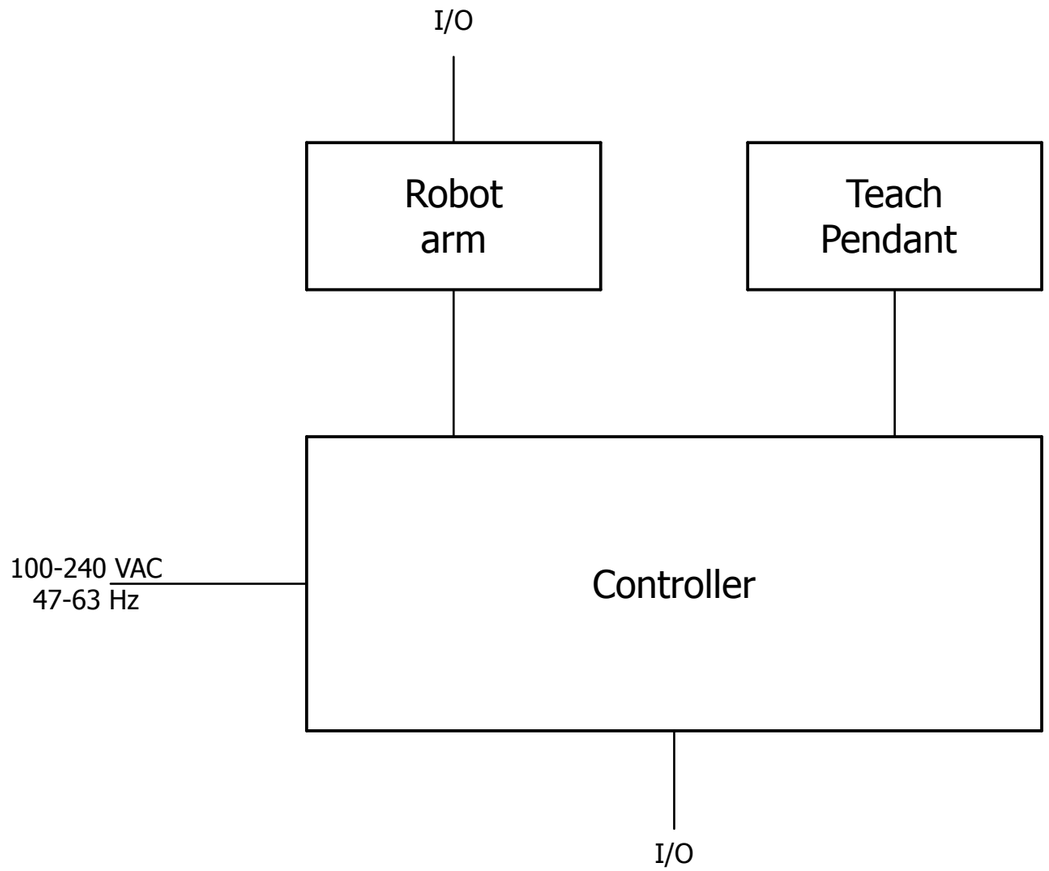
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=UR10 / 23	UR10 Controller I/O		24-11-2015	VKV	
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CB3 Overview UR10



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CB3.0 Controller UR10

Wire colour codes	
Black	BK
Blue	BU
Brown	BN
Green	GN
Orange	OG
Red	RD
White	WH
Yellow	YE
Green/Yellow	GNYE

Supply:
100-240 VAC
47-63 Hz

L
N
GND

CON

Current Distributor

12V PSU: L, N, GND
48V PSU1: L, N, GND
48V PSU2: L, N, GND

BN
BU
GNYE

12V PSU: L, N, GND, +V, -V

48V PSU: L, N, GND, +V, -V

48V PSU: L, N, GND, +V, -V

BK
BK
RD
RD

Energy Eater
+48V
GND

BK
BN
BN
OG
WH
WH
BN
WH
BN
WH
GN
YE
BN
WH
12V

GND
+IN 48V
+OUT 48V
RS485
12V
12V
48V Relay
RS485 12V
TCP/IP

Safety Control Board

12V
SAFETY
ON/OFF
24V
Config. DI
Config. DI
Config. DO
Config. DO
DI
DI
DO
DO
AI/AO
Euromap E67

Compact Flash Card

Motherboard

+ + GND GND
12V
USB
USB
TCP/IP
TCP/IP
DVI
Touch
RS232

GND
8 6 4 7 5 3 1

Robot

RS485
48V
ID=0 Base
ID=1 Shoulder
ID=2 Elbow
ID=3 Wrist 1
ID=4 Wrist 2
ID=5 Wrist 3
Tool
24v DI/DO AI

USB

Teach Pendant

10

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CB3.1 Controller UR10

Wire colour codes	
Black	BK
Blue	BU
Brown	BN
Green	GN
Orange	OG
Red	RD
White	WH
Yellow	YE
Green/Yellow	GNYE

Supply:
100-240 VAC
47-63 Hz

Current Distributor
CON
L
N
GND
12V PSU
48V PSU1
48V PSU2

12V PSU
48V PSU
48V PSU

Energy Eater
+48V
GND

Safety Control Board
SAFETY
12V
ON/OFF
24V
Euromap E67
Config. DI
Config. DI
Config. DO
Config. DO
DI
DI
DO
DO
AI/AO

Motherboard CB3.1
Touch
RS232
DVI
TCP/IP
TCP/IP
USB BLUE
USB
USB
USB
SYSTEM-SW S/N

TCP/IP

Teach Pendant

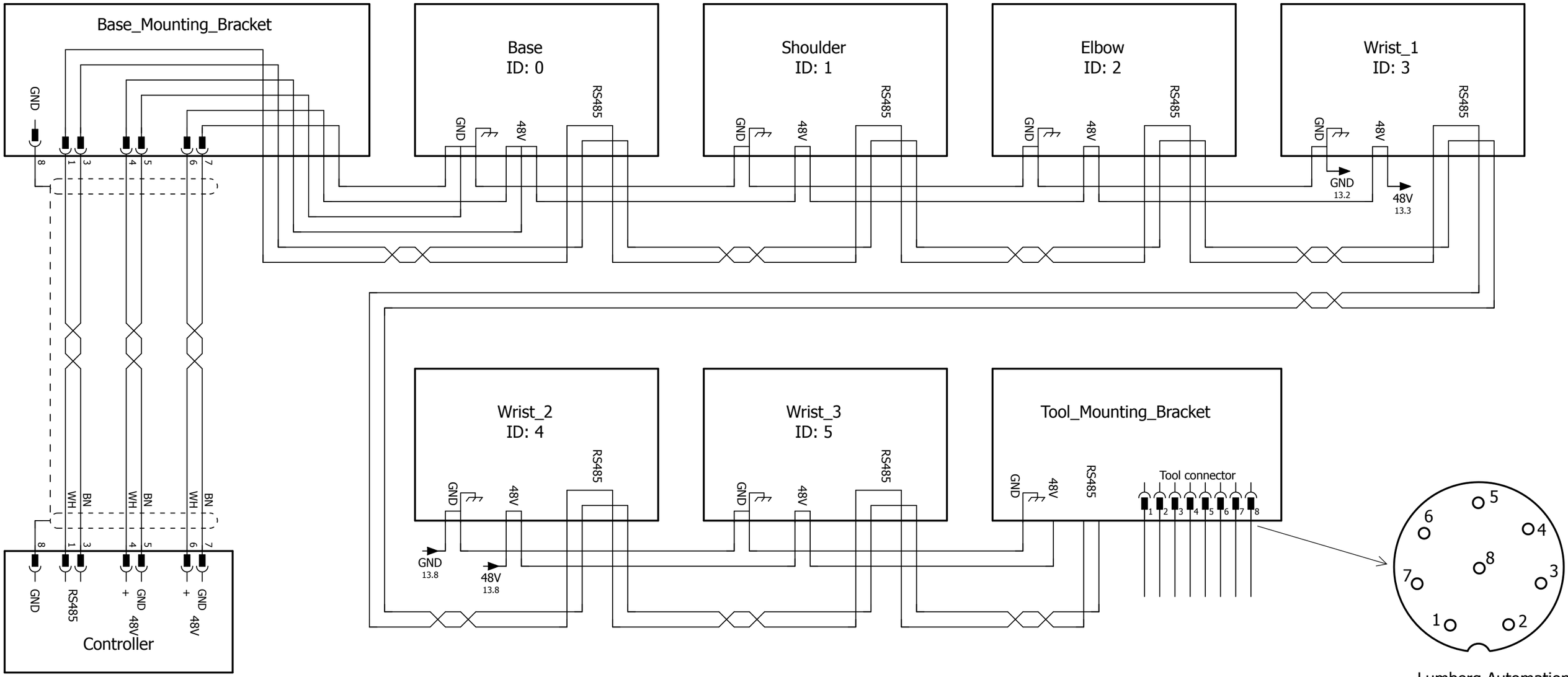
Robot
GND
8 6 4 7 5 3 1
RS485
48V
ID=0 Base
ID=1 Shoulder
ID=2 Elbow
ID=3 Wrist 1
ID=4 Wrist 2
ID=5 Wrist 3
Tool
24v
DI/DO
AI

11

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CB3 UR10 Robot arm



Lumberg Automation
Type: RSMEDG 8
Male Front View

1 = white	AI[2]
2 = brown	AI[3]
3 = green	DI[9] (pnp)
4 = yellow	DI[8] (pnp)
5 = grey	12/24V (power)
6 = pink	DO[9] (npn)
7 = blue	DO[8] (npn)
8 = red	GND

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-UR10
/16.0
/17.0
/18.0
/19.0
/20.0
/21.0
/22.0
/23.0
/24.0
/25.0
/26.0
/27.0

UNIVERSAL ROBOTS

CB3 UR10

L
N
GND

TCP/IP

Supply
100-240 VAC
47-63 Hz

Data I/O

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/15.0

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Safety

Emergency Stop



Default connections

Safeguard Stop



Default connections

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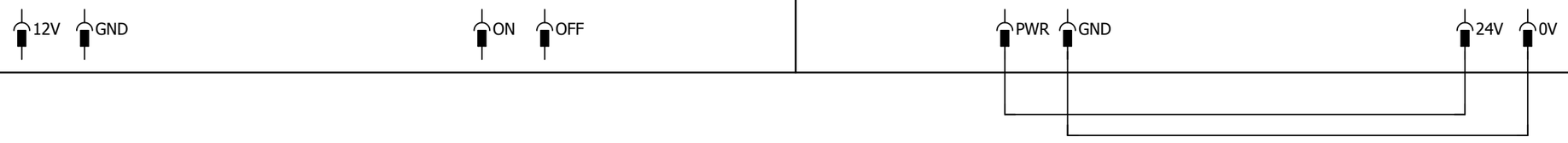
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Remote	Power



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UNIVERSAL ROBOTS CB3 UR10

Configurable Inputs



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UNIVERSAL ROBOTS CB3 UR10

Configurable Inputs



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/15.0

UNIVERSAL ROBOTS CB3 UR10

Configurable Outputs



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-UR10
/15.0

UNIVERSAL ROBOTS CB3 UR10

Configurable Outputs



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UNIVERSAL ROBOTS CB3 UR10

Digital Inputs



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UNIVERSAL ROBOTS CB3 UR10

Digital Inputs



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UNIVERSAL ROBOTS CB3 UR10

Digital Outputs



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Digital Outputs



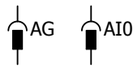
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/15.0

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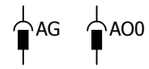
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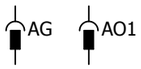
Analog Inputs





Analog Outputs





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Tools connector Inputs/Outputs



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